SAP Cahllenge – Nov 2025

Your name _____ SAP Java

Evaluation:

- The solution is obtained by writing the source code in the given .java file, YourNameExam.java. **You can't add new methods and change the existing ones**
- Each solution must provide a package name in this format ism.ase.ro.sap.exam.lastName.firstName
- **The solution must generate each result on its own. Using hardcoded values** (obtained from other students) **is not allowed** (the solution will be evaluated with 0)
- **To get each requirement points you need to provide an error free (0 compiler errors) program that generates/displays correct results. Incorrect or partially correct solutions are not evaluated**
- All solutions will be cross-checked with MOSS. Solutions that share more than 30% (except the given code) will be evaluated with 0.

You are a forensic cybersecurity specialist employed to recover data from a ransomware attack. This is what you know:

- The ransomware attack has been conducted by a parasitic virus that encrypted most available files.
- The encryption was based on AES, with a 128 bit key, randomly secure generated (you can't brute force it)
- The attack was stopped before the virus had a chance to send the encryption key to the C&C (Command & Control) center.
- The key is stored locally in a random file. From previous investigations you know that is in one of the files from System32 (see the system32.zip given archive – extract the folder).
- Fortunately, you have the SHA2 fingerprint for all those files, computed 1 month ago (before the attack). They are given in the *fingerprints.txt* file. The values are stored in Base64 encoding.

(30 pts) Use the *fingerprints.txt* content to identify the file from system32 folder which has been changed. **(-10 pts if the system32 folder is not placed at the root of the project and you are using absolute paths**)

(30 pts) The ransomware did not use this 16-byte key directly. Instead, it applied a simple obfuscation on each of the 16 bytes to obtain the final AES-128 key. It applied a XOR encryption on each byte with 3Ch and then a cyclic left shift by 1 bit. **Redo the operation** to get the correct binary key. **(-30 points if you hardcode the found filename. At previous step you need to find it and use it)**

(20pts) Using the previously key, decrypt the "*financialdata.enc*" file into "*financialdata.txt*". The virus has encrypted it using AES in CBC mode, with PKCS5Padding. Reverse engineering the virus you find out that that the IV had 1st byte (**from right to left**) equal with 22, 2nd byte equal with 19, 3rd byte equal with 1 and 4th byte equal with 3. The rest of them are all 0s.

(10pts) To confirm your success and get your bounty, write the value of the 1st IBAN into *myresponse.txt* and compute the file HashMAC(based on SHA1). Write the MAC value as Base64 encoded into *myresponse_mac.txt* text file. Don't forget to send the *"financialdata.txt", "myresponse.txt"* and "*myresponse_mac.txt"***.**

Upload

- **the .java file with your solution**
- ***financialdata.txt***
- ***myresponse.txt***
- ***myresponse_mac.txt***